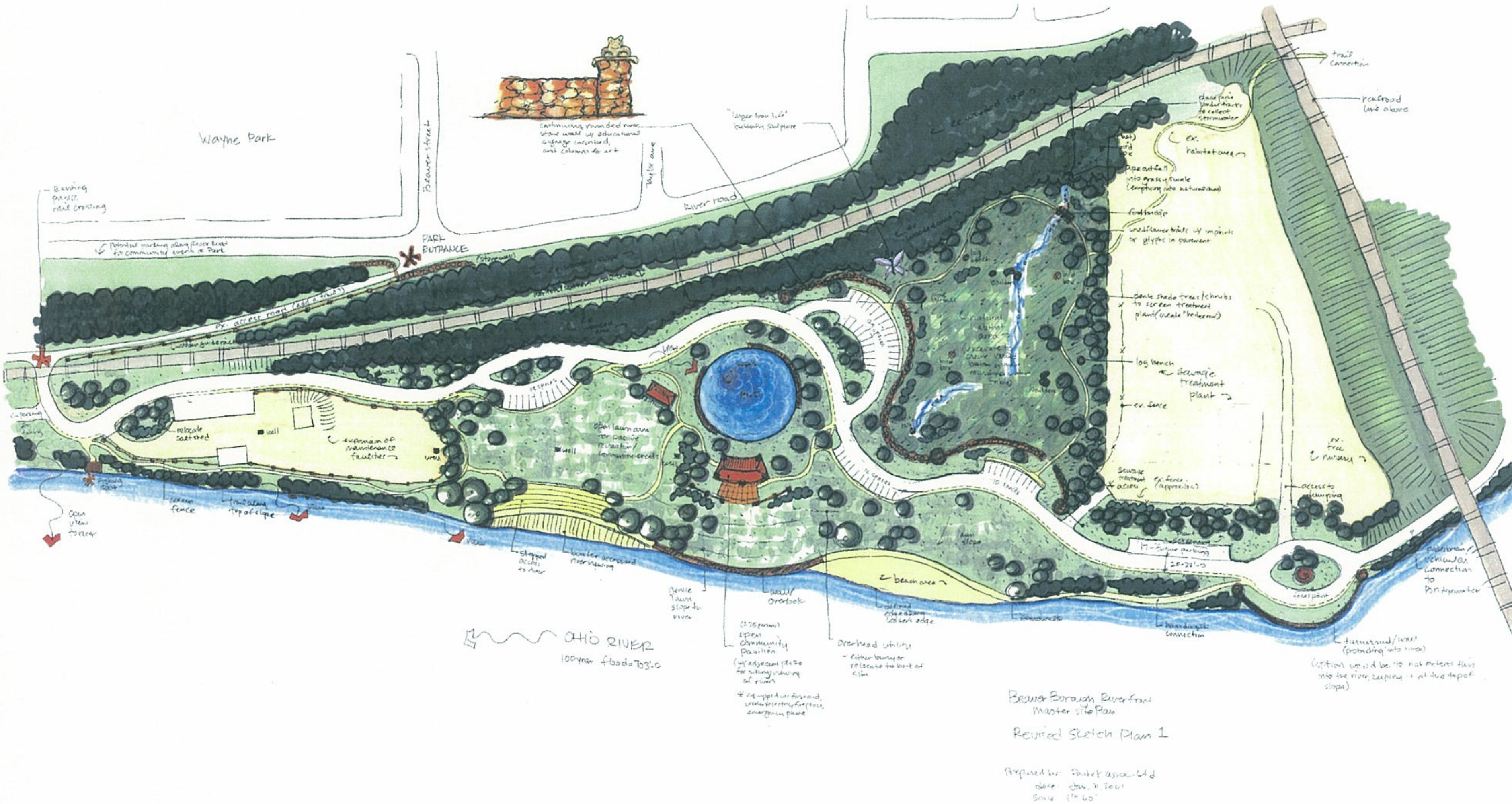
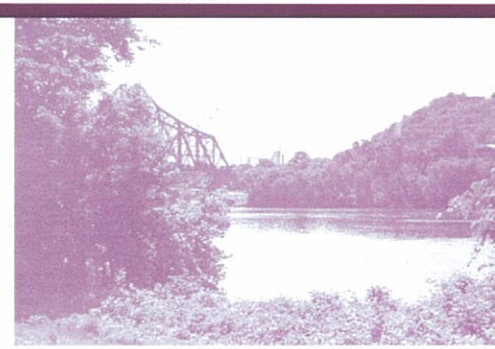
Preliminary Design



Chapter 4

Recommended Design



Chapter 4

Recommended Design

The Beaver Borough Riverfront Park was designed to acknowledge and enhance the nature of the river's riparian environment. In addition, the design is unique to this specific site on the Ohio River. Interpretive elements such as text inscriptions in a river stone wall speak of the story of this local environment's geology and evolution.

The adage, "variety is the spice of life" holds true in the design of recreation facilities. The original goal, to provide recreation opportunities for users of all ages, was clarified and developed by the responses gathered through the public input process. Opportunities for activities such as walking, bicycling, roller-blading, observing nature, viewing the river, playing a pick-up game, or absorbing the calming inspiration of the river are presented in the design.

To successfully integrate the park facilities into the natural environment, an atmosphere was created to encourage recreation, rather than a series of organized recreation elements or objects. A simple park design requiring low maintenance was accomplished by several methods as described in the following paragraphs.

The recommended natural concept and passive recreation design elements for the park will complement other riverfront developments planned for neighboring communities (consistent with the mission of the Beaver County Riverfront Development Plan).

The Riverfront Park will differ from the existing parks in Beaver by offering natural habitat areas in addition to passive open space. To accomplish diversity in experience, a balance of open lawn area to accommodate pick-up games and community events, along with the extension of a natural habitat area was designed. The features of the park center around these two distinct areas.

Community Open Space

Open Lawn Area

A large open lawn area for pick-up games, community events, and other gatherings will be included, rather than structured sports fields. Bordering the open space area, continuous stone sitting walls will be created, with a path along side, rather than individual benches along a trail. Walls create a simple line, withstand flooding, and require minimal maintenance. These walls will also provide a physical barrier for safety, separating the open space activity area from the park access road.

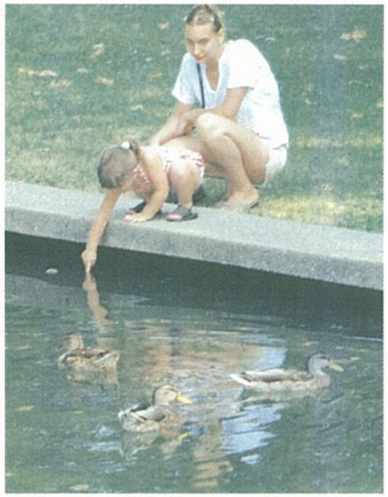
Pond/Water Feature

A water element will be constructed to allow additional recreational opportunities. Such a feature will provide a focal point for the park, allow water access for younger children, provide a place for remote controlled boats, and provide a means for summer wading and winter ice skating.

The 165'-0 diameter pond will be constructed with a concrete bottom. It will be relatively shallow, at 6"-8" in depth, and slightly deeper in the center. It's size was determined by the amount of area available and its proposed use. It was not sized to be a regulation hockey rink, although it can be utilized for recreational hockey pick-up games.

An expansion of wildlife habitat was included to support a broader range of flora and fauna.

The pond will have a concrete edge and be bordered with stone sitting walls, providing a spot to lace up skates or to dangle feet in the water.

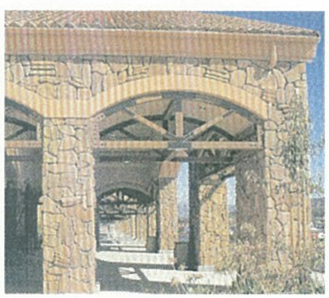


A fountain or spray feature as a focal point and means of water circulation can be included. Water lilies or other vegetation can be placed in the water, in pots, for added summer interest. However, the addition of plants may increase maintenance.

Community Pavilion

A pavilion will be constructed to provide shade, facilitate community gatherings, and serve as a shelter for winter use of the park. It will be architecturally designed and divided to serve large or small gatherings, or a single person relaxing and enjoying the river view. While the shape will most likely be irregular to accommodate these varying uses, it will average 3,600 square feet.

The components of the pavilion will include natural materials, such as stone pillars and a wood shingle roof. Wood detailing accentuating the roof structure, and on other portions of the pavilion above flood elevation, will be included. A durable hardwood should be used (type as suggested in the Appendix). The floor slab will be constructed of exposed aggregate concrete, with a stone terrace adjoining the river side of the pavilion. The pavilion should be designed to discourage bird roosting and nesting opportunities, which creates additional maintenance.



Electrical outlets, an emergency phone, lighting, and water will be included in the pavilion. The structure will be equipped with an integral fireplace and grill, and a water fountain. Lighting around the pavilion will provide flexibility for scheduled evening events.

Restroom

A restroom facility of approximately 25'-0 by 45'-0 will be constructed to serve the users of the park. It will be located central to the main activity areas of the park but be of sufficient distance from the pavilion to minimize notice of odor. The facility will be served by existing utilities (sanitary and water). Underground electric will be installed to the restroom from the relocated overhead lines along the railroad tracks. A water fountain outside the restroom will be included also.

Natural Habitat Area

A sizeable portion of the site will be reserved for the creation and enhancement of natural habitat. An enlargement of the existing natural depression will create the natural area and attract a broader range of fauna. For optimum habitat creation, the area will:

- Be of sufficient size to support wildlife (the proposed habitat area is just under three and one-half acres).
- Have irregular edges with varying sloped grades and banks.
- Have variable, irregular depths to provide seasonal collection of water pockets, with a defined stream channel during wet periods.
- Offer a diverse range of vegetation species, of varying heights and density (Refer to the appendix for suggested grass mixture).
- Provide a buffer around the natural habitat area, providing tree cover.
- Allow surface stormwater to flow through overland swales to filter contaminants and improve water quality.

In addition, large boulders or remnants of old fence posts will be added for wildlife viewing. A curving but continuous rounded river stone wall will provide separation and acknowledge the abundant rounded stone present throughout the Borough. This stone was apparently deposited in the area during the Ice Age, and could possibly have originated from land as far north as the middle of Canada. Educational signage indicating this phenomenon will be inscribed in the stone wall.

The Borough should retain stone uncovered, and stockpile on-site for future use in construction of tab wall. In addition, a community evening or gathering could be assembled to collect stone from residents for future use.

Bird nesting boxes will be placed throughout the natural area to attract a variety of birds. Attracting species such as purple martins and bats will help keep mosquitoes at bay. These birds are attracted to open field areas adjacent to woodlots, with a nearby water source. School environmental clubs can construct the boxes and be responsible for monitoring and maintaining them.

To further set the naturalistic tone, art sculpture will be incorporated at the natural habitat trail entrance. Suggestions for art sculpture include a larger-than-life replica of a butterfly, a sculpture of a frog sitting on a rock, a turtle, or other appropriate wildlife.

Wildlife Attraction

Through careful planning and incorporation of habitat, wildlife should be attracted to the natural habitat area. Species may include many types of birds, such as killdeer, American robin, American kestrel (falcon), mallard, common nighthawk, purple martins, evening grosbeak, Canada goose, little brown bat, ring-necked pheasant, orioles, juncos, downy woodpecker, cardinal, yellow-bellied sapsucker, thrush, bluejay, warbler, titmice, nuthatch, rufous-sided towhee, screech owl, great blue heron, red-winged blackbird, ruby-throated hummingbird, wild turkey, black-capped chickadee, starling, eastern bluebird, and red-tailed hawk. Many of these species will also help to keep pesky insects at bay.

Mammals attracted may include white-tailed deer, eastern chipmunk, woodchuck, short-tailed shrew, muskrat, meadow vole, long-tailed weasel, deer mouse, gray squirrel, beaver, opossum, striped skunk, eastern cottontail rabbit, river otter, and racoon. Proximity to a water source will also be an attracting element, luring frogs, turtles, and snakes as well.

Once established, the natural habitat area may encourage more unusual species such as osprey. The Fish and Game Commission can advise on this possibility. Attracting more unusual species will further the unique character of the site, facilitating its seasonal use as a migration and nesting site.

Community Gardens

Community garden plots for vegetables and flowers will be incorporated into the natural habitat area. Should the gardens become weedy and unkempt, they will blend more readily if surrounded by natural wildflower fields. Crop species should be restricted to annuals, to avoid potential invasive oerennials (i.e., mint) overtaking the native species.

In addition, low, inconspicuous black fencing can be erected to keep ground animals from eating the harvest during the growing season. Once harvesting is complete, a portion of the fence will be temporarily removed to allow wildlife to feast on the remaining food, including vegetables and seeds from flowers.

River's Edge

River Access and Edge Treatment

Along the riverfront, viewing and sitting areas will be created by terracing stone or concrete steps with wide treads, alongside extensive sloped mowed lawn areas. A concrete or stone edge will define a portion of the edge of river to reduce susceptibility to water erosion. In addition, vegetative river bank stabilization will occur along the portion of the river (adjacent to the maintenance facilities), furthering the natural habitat.

A "river walk" will be created along the riverfront, located above the high water level, to help reduce flood damage to the walkway. The walk will be constructed of exposed aggregate concrete or a dense, durable hardwood (refer to the appendix for a suggested material). In some areas due to grade, it may be elevated or stepped with short walls or steps. It will run parallel with the river, acknowledging the contours of the riverbank and being ADA accessible affording visitors views of the river. The walk will proceed up-river, connecting to Bridgewater Crossing, and ultimately linking with the existing (and future proposed) boardwalk at Bridgewater's Bicentennial Park.

A combination of hard edges and green edges are suggested for the river's edge. Hard edges will help curb soil erosion at the riverbank, while green edges will encourage wildlife habitats on both land and water. This combination will promote the health of the river environment. It will also sense to recognize and accept the natural conditions and seasonal cycles of flooding that characterize any river environment.

Boater Access

Boater access to the site will be limited to tie-up locations along the overlook wall, rather than boat docking facilities. This will allow pick-up and drop-off of boaters, and allow boaters access to the park for their enjoyment. In addition, a boater access platform will be constructed near the tie-up spots, with stepped access. This platform can be constructed of concrete or stone, or of a durable hardwood (type as suggested in the appendix), elevated approximately two feet above the water level.

Beach

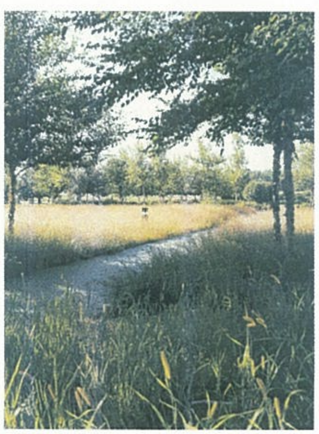
A sandy beach along a section of shoreline will be constructed, using a base of larger rock choked with a finer sand or pebble. Should the finer pebble be washed away with flooding, the larger stone base will remain, thus reducing maintenance.

To further reduce maintenance frequency, the beach areas should be constructed above the normal pool elevation. A concrete edge will be constructed to contain the beach, holding it above this normal pool elevation.

Fishing Platform

An accessible platform for fishing will be constructed near the existing sewer outfall along the riverbank. Sightings of great blue heron tell us fish are attracted here. The platform will be constructed of a durable hardwood and be of size to accommodate up to five persons. An accessible trail will join the adjacent parking area with the fishing platform.

Connecting Elements



Trails

Pedestrian trails will be constructed parallel to the access road, to accommodate walking, biking, and roller-blading through the site and to adjoining areas. Constructed of bituminous pavement, these ten foot wide multi-use trails will be separated from the road by a lawn strip of varying width. Walking trails will be accessible, with slopes less than five percent, to meet the Americans with Disabilities Act (ADA) guidelines.

Physical connections of this park to the adjacent Bridgewater Crossing will occur at two additional locations: via the arched stone railroad tunnel off the trail in the natural habitat area, and along the water's edge via a river walk. This latter connection will eventually join the existing (and future proposed) boardwalk at Bridgewater's Bicentennial Park.

Pedestrian trails through the natural habitat area will be restricted to walking, and located at the edges to reduce impact on wildlife. Constructed of crushed stone to encourage a slower pace, these accessible trails will be five feet in width. They will be a natural stone with glyphs or imprints of wildlife footprints in the surface. Occasional sitting areas will be provided and constructed of large logs, rather than benches, in keeping with the natural theme. Trails through the natural habitat area will allow the visitor enjoyment of wildlife viewing and the opportunity to spot native wildflowers.

Visual Connections

Enhancing visual connectivity will be accomplished through further opening views to the river and elements beyond, such as the architecturally significant railroad bridge. This can occur at the river's edge by selective removal of lesser trees and underbrush, rather than healthy, mature specimen trees. Invasive species, such as Japanese Knotweed, will be effectively removed by excavation of its roots, and repeated until complete removal is accomplished.

Views to the park from River Road can be enhanced in similar ways. Maintaining selective clearing of trees and underbrush on the slope separating the site from the town of Beaver, as well as performing selective clearing of trees between the railroad tracks and the site, will allow more open views to the park, the river, and those elements beyond.

Art in Design

The incorporation of art into the design of the park will also contribute to a unique sense of place. Design competitions can be initiated, whether at the professional, amateur, or student level to elicit creative solutions and encourage public participation in the park design.

Different forms of artistic expression can be incorporated, such as sculpture, earthen forms, or display gardens. Art can be incorporated into the design of the community pavilion. One suggestion involves a frog on the roof, with a leg or two hanging down over the edge. Art sculpture can also be incorporated into the natural habitat area, as described previously.

A focal point or other feature such as a light house at the park road turn-around can be an artistic expression which identifies Beaver to those utilizing the river. The art should enhance the overall vision of the park and provide visitors with a means to relate to the environment

Educational Opportunities

The Beaver Riverfront Park will support environmental education activities such as habitat creation, nesting monitoring, and water quality studies. Opportunities exist for access to educational programs, such as "River-Ventures" through the Pittsburgh Voyager. Voyager is a non-profit river-based organization that provides high quality environmental science, river ecology, and mathematics education to students in Western Pennsylvania.

The construction of bird nesting boxes, butterfly nesting boxes, bat boxes, and other shelters can be completed through school programs.

Birding is a popular recreational and educational activity which will be afforded in the natural habitat areas of the park, as well as along the riverfront. Numerous birds of various species currently frequent the site, and more are anticipated with the enhanced habitat.

Plantings

Plantings for the site should be native to the riverfront area, where feasible, or to the state of Pennsylvania. This will avoid competition with introduced species, which usually offer less of a food source for wildlife. A native plant can be defined as a plant occurring naturally in a region, ecosystem or habitat. These plants help promote regional identity, and enhance bio-diversity.

Large trees for shade and wildlife shelter will be planted in random fashion through the park and natural habitat area. They will be clustered more closely, in a natural pattern along the Beaver/Bridgewater property line, to effectively screen the sewage treatment plant. A mixture of understory flowering trees, shrubs, and field grasses, with wildflowers, will be added to enhance the beauty and wildlife habitat, providing appropriate food and shelter.

Shrub species such as gray or silky dogwood, hawthorn, alder, and willow enhance wildlife cover. Alder and legume species of grasses have the added benefit of increasing earthworm production and soil fertility through nitrogen fixation.

Specific plantings to attract and sustain various forms of wildlife should be targeted for planting. Those plantings which offer fruit, nuts, and sufficient cover are most important. Seasonal crops can be planted, or be left over from community gardens to supplement food sources.

Native non-mowed grass species (a combination of warm season and cool season grasses) will comprise the base of the natural habitat area. Native perennials will then be planted in random fashion. These grassy herbaceous fields will be mowed only annually, avoiding late spring or early summer since this represents the nesting period for grassland birds. Similarly, fall mowing of fields should be avoided, as the grasses provide food for such species as the monarch butterfly. Migrating butterflies feed in the larva stage, then go into a period of change while still using the plant on which they fed. Mowing early in the fall will destroy these larva and interrupt the cycle of the butterfly.

The river stone wall provides a physical reminder of the limits of the natural habitat area, and its reduced mowing requirement. Regularly mowed lawn areas will be reserved for the open space area, and the lawn slopes along the river for sun bathing or viewing. In addition, a mowed lawn strip will occur between a trail where it occurs along the road and the natural habitat area.

Infrastructure

Park Entrance

At the park entrance on River Road (at Beaver Street) and at the park exit (at Market Street), stone walls with a park sign and wood swing gate will be constructed on either side of the road. Attractive entrance plantings will also be added to announce the park.

Vehicular Access

Main vehicular and pedestrian access to the site will occur via the current park entrance on River Road at Beaver Street. The one-lane access roads leading to and from the public railroad crossing will be upgraded to improve safety, and will include raising storm inlets to be flush with the road surface. These road services will be overlayed or re-paved with bituminous. Along each one-way access road a wooden guide rail (PennDot approved) will be placed along the downslope edge, for safety. Where feasible and slope allows, a walking lane will be created along each edge.

Roadways within the park are proposed with a drive aisle width of twenty-two feet, at ten miles per hour. Reduced pavement widths will minimize visual impact of paved areas and reduce stormwater runoff. Two-way roads are suggested, mainly to avoid traffic control signs cluttering the site.

To improve safety at the track crossing, the access road will be aligned to a more straight position on the river side of the tracks. This will provide better sight distance and access and take advantage of the river view. The salt storage shed will need to be relocated to accommodate this realignment.

The road will curve back to traverse between the maintenance buildings and the track (following the existing road), and proceed straight past the water treatment plant, eliminating the curve and poor sight distance of the existing road in this area. The road will then continue along the back side of the tracks before turning towards the river.

The site access road will end beyond the site boundary, on the river side of the wastewater treatment plant. This termination will be a circular roadway turn-around, projected out into the river to maximize views of this area. The design includes a large focal point at the center of this turn-around, visible to both land and water visitors. The focal point will be an art sculpture, monument, or lighthouse created by the community.

A road connection from the turn-around will proceed under the railroad bridge, to the future Bridgewater Crossing. Access to the sewage treatment plant and Borough yard waste and dumping facility will also be maintained in this area.

Railroad Track Crossing

At the convergence of the two site access lanes exists a public railroad crossing. This crossing has been verified to be a public crossing (refer to the appendix for verification from Norfolk Southern Corporation). The crossing will be upgraded by re-paving the surface, and installing cross-arms and signage (as coordinated with Norfolk Southern Corporation).

Parking

Approximately 64 parking spaces are initially proposed to support park facilities. These will be constructed in locations about the site so they are effectively utilized and to minimize parking in non-designated areas. The parking areas will be constructed of bituminous paving, and will be 9'-0 feet wide by 18'-0 feet long. For large community events, overflow parking can occur off-site along River Road, where current parking is allowable. Future parking to accommodate 17 spaces is allocated near the park access road turn-around, and will only be constructed if necessary.

Parking Analysis

FACILITY	Capacity (in persons)	Parking Required
Community Pavilion	100	50
Pond	25	15
Habitat Area	15	8
Fishing Pier	8	5
Peak Use Totals	148	78
Daily Use Neet (60% of peak)	47	
Spaced Provided	64	
Future Parking Spaces	17	
Total Spaces Available	81	

Stormwater Management

The handling of stormwater generated from new site improvements, as well as existing runoff entering from the town of Beaver, is proposed to be handled through on-site measures. The surface stormwater discharge opposite Taylor Avenue on River Road collects surface water from a portion of the town of Beaver, and currently exits into the site. In addition, stormwater collecting on the north side of the tracks (below Wilson and Lincoln Avenues) can be routed under the tracks (through a pipe) and to the site.

Each of these outlet points will be collected via surface swales, or underground piping, then routed through the natural habitat area. This will allow further filtering of the runoff before it enters the river. Another benefit to handling surface stormwater in this fashion will be a greater range of habitat and attraction of wildlife to the natural habitat area, through the addition of this water. The number of stormwater inlets on the site will be minimized, where feasible.

The implementation of these stormwater management concepts are consistent with the Best Management Practices supported by the PADEP, and with Growing Greener Bio-Engineering Solutions.

Maintenance Facilities and Access

For the near future Borough maintenance facilities will remain in their current location. The salt storage shed will be replaced, and relocated (preferably off-site) to allow better entrance circulation and to avoid having to relocate salt piles to higher ground prior to anticipated flooding. Eventually, the Borough should pursue off-site relocation of all maintenance facilities (sheds, laydown areas, etc.) to restore the site to its original state and to allow more recreational opportunities at the river's edge. The site should be reserved for public recreational use.

A fence enclosing the maintenance facilities will be constructed to offer separation and discourage entry into this area. The fence will be constructed of materials to withstand occasional flooding, such as steel posts and mesh, rather than wood. Planting can be incorporated to provide additional visual screening. It is also recommended the structures within the maintenance facilities be painted a neutral color to blend more readily with the surrounding environment.

Access to the resident yard waste area, behind the wastewater treatment plant, will be maintained via the park access road. The Borough may pursue an alternate, off-site location.

Signage

Limited signage will be designed and constructed of natural materials and subdued colors, in keeping with a natural theme of materials for hardscape elements. Required signage might include park entrance sign(s), accessible parking, park rules, and educational signage.

Utilities

The overhead electric utility line (running along the existing site road) will be either relocated underground, or be routed along the railroad tracks. This will help to reduce visual disruption of river views.

The depths of the remaining underground utilities within the site will need to be determined prior to setting proposed grades. Connections to these utilities, including sanitary and water, will provide service to the proposed facilities.

Lighting is proposed to aid in surveillance and to provide illumination for night-time scheduled events. Lighting will be concentrated along the park access road, and in the area of the community pavilion and pond. Lighting should be designed to illuminate the park without inducing glare beyond the activity area. Glare should be directed away from the residential district above the site, and not spill over onto the river corridor.

Remaining Items

Naming of the Park

Consideration should be given to naming the park to further the sense of place, making it unique to this site and community. Ideas are as follows:

- Great Path Park (acknowledging The Great Path, the main westward migration mode)
- Old Town River Park or Beavertown Park (acknowledging the former name of the town of Beaver)
- Shingas Park (acknowledging the Indian leader Shingas)

Vandalism

Vandalism concerns will be reduced with an increase in visitor frequency. By providing a variety of activities, the park will be utilized at random times of the day. Provision of many outdoor activities, such as walking and biking trails, and access to the water for fishing, also contributes to expanding lifestyle choices for current and new residents in the area.

Existing Water Wells

There is some concern regarding the construction of park improvements on a site which contains the Borough's water wells. Discussions were held with the study committee and representatives of the Borough. The following is a summary of issues identified:

1. Location of proposed improvements may need to avoid the cone of depression around each well source. A cone of depression study will need to be completed to determine if this is a limiting factor.

2. Large truck or crane access should be provided to each well source for periodic monitoring and maintenance.

3. Consider construction of a removable covering over the wells to disguise their views. The wells can be equipped with an alarm system, connected to the police station.

4. Drilling of future new wells should occur in locations beyond proposed improvements, when feasible.

In addition, the consultant posted a query on the National Recreation and Parks Society Association (NRPA) website to obtain practical information from other municipalities around the country regarding this issue. A summary of these responses can be found in the appendix.

Acceptable Facility Standards

The design of the Beaver Borough Riverfront Park incorporates acceptable facility standards as defined by the following organizations:

- Americans with Disabilities Act, Title II, Requirements for Public Facilities; requires public facilities be accessible to the physically challenged. Includes proposed structures, trails (preferably all, or a portion thereof), parking, and access to the river.

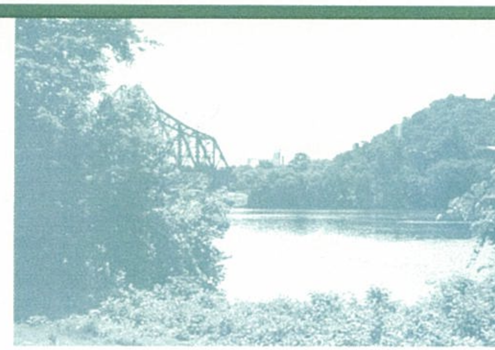
- National Park and Recreation Association's Facility Development Standards; for general facility construction.

- American Association of Highway and Transportation Officials Guide for the Development of Bicycle Facilities; for bicycle trails.

Recommended Plan

Chapter 5

Implementation



Chapter 5
Implementation

Estimate of Development Costs

An estimate of costs was prepared for the recommended design of the Beaver Borough Riverfront Park. This estimate was based on the assumption that implementation of park development will occur through a public bidding process. Costs reflect year 2001 wage rates. In Pennsylvania, all projects over $25,000 are required to use the State's Prevailing Wage Rates for construction. To budget for inflation of costs for phases beyond Phase 1, include a four percent annual adjustment.

With volunteer labor and donated materials, the costs associated with these park improvements may be less than estimated. Additionally, alternate sources of funding, including grant opportunities as described herein, may help defray direct development costs.

Permitting and Approvals

Given the presence of hydric soils, a delineation and determination of potential jurisdictional wetlands will be required prior to implementation of site improvements. Should wetlands be identified, they will require protection or replacement.

Other necessary permitting may involve the Pennsylvania Department of Environmental Projection (for erosion and sedimentation controls), the Pennsylvania Fish and Boat Commission, and the Corps of Engineers. Coordination with Norfolk Southern Corporation will need to occur regarding upgrade of the public track crossing, with approval for proposed work shown within a portion of their property. It should be noted that portions of the existing site road is within railroad property.

Construction Projects

The future development of the park may involve various types of construction projects, which may impact the material or labor cost of improvements in each phase. The following is a brief summary of the suggested types of construction projects that may be undertaken:

<u>Contracted Projects</u>

Beach
Entrance/exit walls, gates, and signage
Fence installation
Lighting
Overhead electric relocation
Pavilion and terrace
Pond with walls
Restroom
River access (river walk, steps, and walls)
Road and parking bituminous paving
Site preparation and grading
Wood guiderails

<u>Public Works Projects</u>

Fence
Fishing platform
Manhole and inlet elevation adjustment
Road bed and parking lot base
Salt storage shed relocation
Sanitary sewer lines
Stormwater lines
Trails
Water lines; fire hydrant

An expansion of wildlife habitat was included to support a broader range of flora and fauna.

<u>Service Projects</u>

Art sculpture and focal points
Boulders in habitat area
Community gardens
Crushed stone trails
Footbridge
Lawn and field seeding
Log Benches
Nesting boxes and bird houses
Planting

Phased Capital Improvement Plan

Given the estimated total cost, park improvements have been phased to reflect an anticipated and logical progression of construction. A description of these anticipated phases and their associated costs are described herein. Per the requirements of DCNR, the total estimated costs reflect prices as if each phase was a contracted project.

Phase I: Safe Site Access ($504,400)

The first phase of park development will be the upgrade of the existing one-lane access roads serving the site, to include the installation of wood guide rails and overlaying the bituminous surface. To provide an uninterrupted surface, the existing stormwater inlets along these roads should be raised flush with the new overlay surface. In addition, upgrade of the public railroad crossing is recommended via crossing arms, signage, and re-surfacing. These improvements will serve to provide safe access prior to promoting the site for public recreational use.

Phase II: Site Preparation ($540,675)

The second phase of park development will include preparing the site to receive proposed design elements. This will include brush hogging and selective tree removal, pruning of trees to remain (at riverbank), existing asphalt road removal, relocation of the overhead electric line, lowering sanitary manhole top elevations, overall site grading, storm water drainage (collection and redirection of Taylor Avenue drainage), and site stabilization through the seeding of lawn, and of field grasses in the habitat area.

Phase III: Initial Site Improvements ($724,000)

The third phase will include construction of initial site improvements, such as site access roads and parking, the turnaround with focal point (and wall projected into river), associated lighting, and directional signage. A multi-purpose trail will be constructed parallel to the road. The salt storage shed will be replaced (preferably at an off-site location, or temporarily behind the wastewater treatment plant) to allow a safer alignment of the access road in this area. Construction of the fence enclosing the existing maintenance area will occur at this time, providing separation and discouraging trespass. Trash receptacles will be provided to discourage littering.

In addition, the park entrance and exit features, including stone walls, entrance gates, signage, and planting will occur in this phase. A fire hydrant will also be installed.

Phase IV: River's Edge Improvements and Natural Habitat ($732,350)

The fourth phase will include the river's edge improvements. The boater access and river viewing platform, the stepped access, the stone river overlook wall and columns, the river walk, walking trails (to the overlook wall and stepped access areas), and the beach will be included. Trees will also be planted in this area for shade and scale.

In addition, this phase will include the completion of the natural habitat area, with the defining river stone wall with educational inscriptions, nesting boxes, art sculpture, boulders, log benches, crushed (natural stone) walking trail, and footbridge. This phase will also include the optional collection and direction of off-site stormwater from Lincoln Avenue, under the railroad tracks, into the natural habitat area.

To effectively drain the habitat area, an outlet structure connected to a storm pipe will be installed, outletting to the river through the overlook wall. A flap will be installed at this pipe end to prevent back flow of river water into the pipe during high water fluctuations.

Phase V: Restroom ($280,975)

The fifth phase will include construction of the restroom building and associated utilities. In addition, the stone sitting walls at the open space lawn area will be built. In this phase, the existing maintenance buildings will be painted a subdued color to blend more readily. A trail (for biking and walking) will be constructed at the top of the riverbank, connecting the fishing platform to the main park area.

Phase VI: Community Pavilion ($505,750)

The sixth phase will include construction of the community pavilion and associated utilities. The stone patio terrace and flanking stone walls will be constructed, with associated site lighting. A walking trail connecting the pavilion to the parking area will be constructed. Trees and shrubs will be planted for shade and color.

Phase VII: Pond / Water Feature ($279,975)

The seventh phase will include construction of the pond/water feature, spray/bubbler, lights, adjacent stone sitting walls, and associated utilities. Trees and shrubs will also be planted.

Funding Sources for Proposed Park Improvements

Many agencies provide grants to assist in providing financial resources to implement the design and construction of parks and recreation facilities. Some offer grants to implement educational programs in concert with these facilities. Still others support the planning and implementation of projects which preserve habitat and improve watersheds. Assistance can also take the form of technical help, information exchange, and training.

Strategies

Submission of a thorough application may result in award of monies, given the competition for grant funding. Strategies for improving the chances of receiving a grant include:

- Being well-prepared by knowing the funding agency (contact persons, addresses, phone numbers); ensuring your municipality and the project are eligible; and submitting a complete and accurate application ahead of the deadline.
- Clearly indicating the Park's Vision and Plans in the application, to portray where your project fits into the community. Describe how matching funds such as local taxes, private contributions, and other grants will leverage the funding. Describe how maintenance of the park will be accomplished, to help justify the request for the grant. Show past successes with Borough and recreation being funded and built, and how this project impacts those successes.
- Contacting the funding agencies by personally meeting with them to show your commitment to the project.

Additional strategies for successfully obtaining grant monies is included in the appendix.

Funding Opportunities

The Appendix holds a multitude of sources for obtaining potential grants to fund park improvements. These sources are described in detail, including contract information. Below are those which hold the most likelihood of success:

<u>Community Conservation Partnerships Program - Acquisition and Development Grants - Community Grants Program</u>

Through the Pennsylvania Department of Conservation and Natural Resources (DCNR), this grant program supports land acquisition, and park rehabilitation and development for small communities.

<u>Single Application Grants</u>

Through Pennsylvania's Center for Local Government Services, Department of Community and Economic Development, funding can be obtained for a wide variety of municipal projects, including recreational facility improvements and development.

<u>Community Conservation Partnerships Program - Pennsylvania Recreational Trails</u>

Through DCNR, in consultation with the Pennsylvania Recreational Trails Advisory Board, this program provides funds to develop and maintain recreational trails.

<u>Pennsylvania Conservation Corps</u>

Through the Pennsylvania Department of Labor and Industry, this program provides work experience, job training, and educational opportunities to young adults while accomplishing work on public lands.

<u>Surface Transportation Program (STP) Funds</u>

Through the Federal Highway Administration of the Department of Transportation, available funding can be used for bicycle and pedestrian facility construction or for brochures and route maps. May include funds for upgrades of railroad crossings.

<u>Transportation Equity Act for the 21st Century (TEA21)</u>

Provides the primary source of federal funding for greenways and trails through the Transportation Equity Act of 1998.